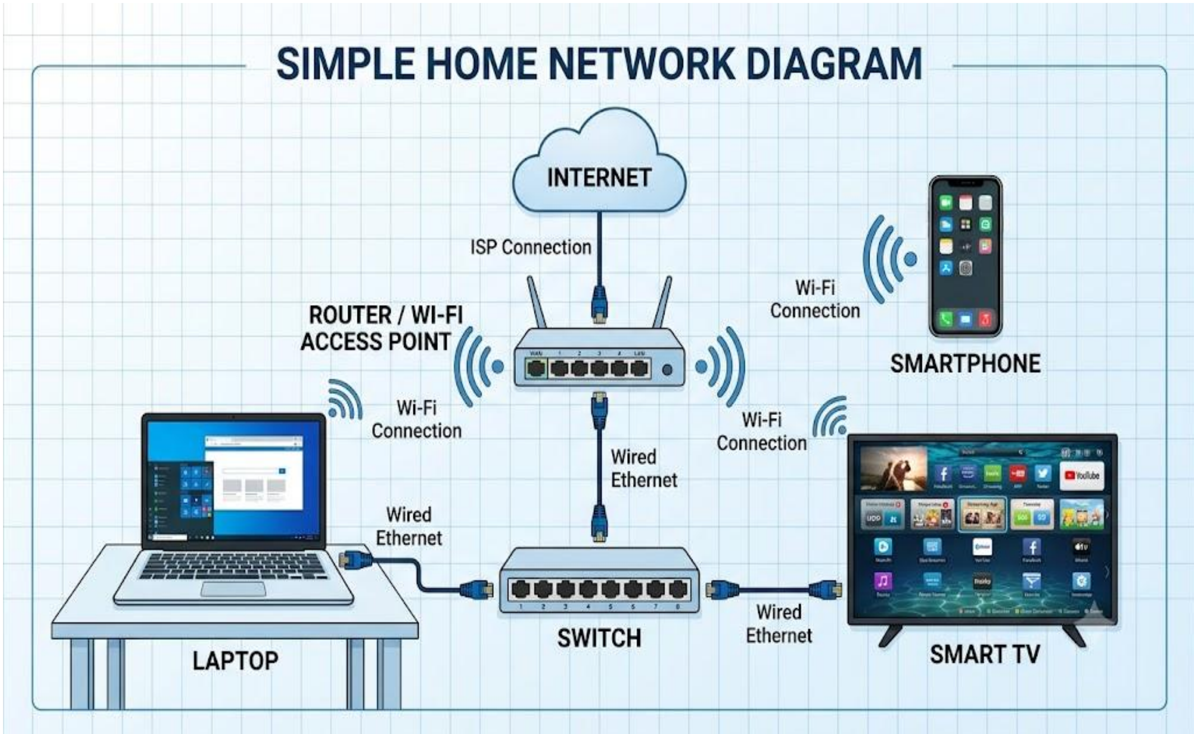
Network devices

Task 1 :- Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.

ANS :-



Task 2 :- Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

ANS :-

Device	**Primary Role**	**Real-World Daily Use Case**
Router	Connects your home network to the Internet and assigns local IP addresses.	Streaming 4K videos on YouTube or Netflix over your home internet connection.
Switch	Connects multiple wired devices together on the same local network.	Setting up low-latency LAN gaming (like *Counter-Strike/ Valorant*) between multiple gaming PCs.
Hub	Broadcasts incoming data to *all* connected ports simultaneously (legacy hardware).	Joining a simple retro multiplayer match or monitoring network traffic in basic lab setups.
Access Point (AP)	Extends wireless signals to give devices Wi-Fi access.	Scrolling Instagram Reels or making WhatsApp video calls on a smartphone in a room far from the main router.

Task 3 :- Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

ANS :-

1.Smartphone (Device) :

You tap "Place Order" on the Zomato app. Your phone sends the order data out as radio waves.

2.Access Point (AP) :

Catches those radio waves from your phone and turns them into wired electrical signals.

3.Switch :

Takes the wired signal from the AP and directs it straight to the router without sending it to unrelated local devices.

4.Router :

Reads the destination address of your Zomato request and sends the data package off into the Internet, connecting to Zomato's servers to confirm your order.

Task 4 :- Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.

ANS :-

1. **Powerline Adapter Set :-**

- How it works: You plug one adapter into a wall outlet near your router and connect it to the router using a short Ethernet cable. You plug second adapter into a wall outlet near your PC and connect an Ethernet cable from it straight into your PC.

- The Magic: It turns home's existing electrical wiring into a fast, wired network cable.

2. Wi-Fi Extender (with an Ethernet port) :-

- How it works: You plug the extender into a wall outlet roughly halfway between your router and your PC where it can still get a good Wi-Fi signal. The extender receives the Wi-Fi wirelessly from the router, and you plug an Ethernet cable from the extender's built-in Ethernet port directly into your PC.

- The Magic: It acts as a wireless bridge, catching the Wi-Fi signal through the air and converting it into a wired Ethernet connection for your PC.